

Arithmetic in Spark `hy_rk_correctFluxes`

Source: `source/physics/Hydro/HydroMain/Spark/hy_rk_correctFluxes.F90`, including its internal routines `correctZone` and `geoFacs`. The routine applies supplied flux differences to boundary cells and calls the equation of state after each correction.

1 Notation and supplied flux differences

Let $q = (i, j, k)$ denote a cell, $h_d = \text{del}(d)$ the block spacing in direction d , and $\Delta t = \text{dt}$. The cell state contains density ρ , the three velocity components $\mathbf{v}$, specific stored energy $e = \text{ENER_VAR}$, specific internal energy ϵ , and species/mass-scalar values X_s . For MHD, it also contains $\mathbf{B}$ and $\psi = \text{PSIB_VAR}$. All three vector components are used in the energy and momentum arithmetic, even in lower-dimensional builds.

Write the supplied buffer on a face as $\delta \mathbf{F}_d$. The source describes these buffers on the coarse side of a fine/coarse interface as

$$\delta \mathbf{F}_d = \sum \mathbf{F}_{d,\text{fine}} - \mathbf{F}_{d,\text{coarse}}, \quad \delta \mathbf{F}_d = \mathbf{0} \text{ elsewhere.} \quad (1)$$

The summation, averaging, and any prior weighting of these buffers are performed outside this routine; this expression records the source comment. The code visits every block boundary, so the caller must supply zero buffers where correction is unnecessary.

2 Boundary signs and geometry factors

For each active direction, let $-$ and $+$ denote the low and high sides. The signed vector passed to `correctZone` is

$$\mathbf{D}_{d,-} = -\delta \mathbf{F}_{d,-}, \quad \mathbf{D}_{d,+} = +\delta \mathbf{F}_{d,+}. \quad (2)$$

Specifically, the low-side buffers use face indices i, j , or k ; the high-side buffers use $i + 1, j + 1$, or $k + 1$, respectively.

For Cartesian geometry, $f_{x,-} = f_{x,+} = 1$. For non-Cartesian geometry, write V_q for the cell volume, $A_{x,-} = \text{faceAreas}(i, j, k)$ and $A_{x,+} = \text{faceAreas}(i + 1, j, k)$. Then

$$f_{x,-} = \frac{A_{x,-} h_x}{V_q}, \quad f_{x,+} = \frac{A_{x,+} h_x}{V_q}. \quad (3)$$

If $\text{xCenter}(i) < 0$, the code replaces both factors with zero. In the y and z boundary corrections, the code always sets

$$f_{y,-} = f_{y,+} = f_{z,-} = f_{z,+} = 1. \quad (4)$$

Only x uses the face-area/volume geometry factors in this routine. Face areas and volumes are obtained from Grid calls; their geometry-specific formulas are not implemented here.

3 Primitive-to-conserved conversion

Each individual boundary update reconstructs a conserved vector $\mathbf{Q}$ from the cell's current primitive state:

$$Q_\rho = \rho, \tag{5}$$

$$\mathbf{Q}_m = \rho \mathbf{v}, \tag{6}$$

$$Q_E = \rho e, \tag{7}$$

$$Q_s = \rho X_s \quad \text{for each species/mass scalar } s. \tag{8}$$

When `SPARK_GLM` is defined, the additional assignments are

$$\mathbf{Q}_B = \mathbf{B}, \tag{9}$$

$$Q_\psi = \psi, \tag{10}$$

$$Q_E = \rho e + \frac{1}{2} \mathbf{B} \cdot \mathbf{B} = \rho e + \frac{1}{2} (B_x^2 + B_y^2 + B_z^2). \tag{11}$$

Thus the stored specific energy is converted to an energy density, and magnetic energy density is added separately. No ψ energy term is added.

4 Conserved-vector correction

For one side in direction d , the entire conserved vector is updated by

$$\boxed{\mathbf{Q}' = \mathbf{Q} - \frac{\Delta t}{h_d} f_{d,\pm} \mathbf{D}_{d,\pm}.} \tag{12}$$

Using the unsigned supplied buffers makes the side signs explicit:

$$\mathbf{Q}'_- = \mathbf{Q} + \frac{\Delta t}{h_d} f_{d,-} \delta \mathbf{F}_{d,-}, \tag{13}$$

$$\mathbf{Q}'_+ = \mathbf{Q} - \frac{\Delta t}{h_d} f_{d,+} \delta \mathbf{F}_{d,+}. \tag{14}$$

For non-Cartesian x corrections at nonnegative radial centers, the coefficient simplifies to

$$\frac{\Delta t}{h_x} f_{x,\pm} = \frac{\Delta t}{V_q} \frac{A_{x,\pm}}{V_q}. \tag{15}$$

These formulas apply componentwise to mass, momentum, energy, species/mass scalars, and (in GLM-MHD) magnetic field and ψ .

5 Conserved-to-primitive conversion and floors

For GLM-MHD, the magnetic and cleaning fields are recovered directly:

$$\mathbf{B}' = \mathbf{Q}'_B, \quad \psi' = Q'_\psi, \tag{16}$$

$$E'_B = \frac{1}{2} \mathbf{B}' \cdot \mathbf{B}', \quad \text{MAGP_VAR}' = E'_B. \tag{17}$$

The energy density used to recover the stored specific energy is

$$E'_{\text{fluid}} = \begin{cases} Q'_E, & \text{without SPARK_GLM,} \\ Q'_E - E'_B, & \text{with SPARK_GLM.} \end{cases} \quad (18)$$

Let $\rho_{\min} = \text{hy_smallldens}$ and $e_{\min} = \text{hy_smallE}$. The remaining assignments are

$$\rho' = \max(Q'_\rho, \rho_{\min}), \quad (19)$$

$$\mathbf{v}' = \frac{\mathbf{Q}'_m}{\rho'}, \quad (20)$$

$$e' = \max\left(e_{\min}, \frac{E'_{\text{fluid}}}{\rho'}\right), \quad (21)$$

$$e'_{\text{kin}} = \frac{1}{2} \mathbf{v}' \cdot \mathbf{v}' = \frac{1}{2} ((v'_x)^2 + (v'_y)^2 + (v'_z)^2), \quad (22)$$

$$\epsilon' = \max(e_{\min}, e' - e'_{\text{kin}}), \quad (23)$$

$$X'_s = \frac{Q'_s}{\rho'}. \quad (24)$$

The density floor is used in every division. The routine does not clip or renormalize X'_s . The internal-energy floor is assigned independently; this assignment does not recompute e' as $\epsilon' + e'_{\text{kin}}$. Consequently, the floors can alter the exact conserved-to-primitive correspondence.

6 Equation of state and update order

After every boundary-side correction, the code invokes `Eos_getData`, `Eos(MODE_DENS_EI)`, and `Eos_putData` for that cell. Symbolically,

$$\text{cell state after EOS} = \mathcal{E}_{\rho, \epsilon}(\text{corrected cell state}). \quad (25)$$

The EOS closure is external to this file, so no particular pressure or temperature formula is inferred here.

The routine processes x sides first, then y sides for $\text{NDIM} > 1$, then z sides for $\text{NDIM} = 3$. Each transverse-index iteration processes the low side before the high side. Edge and corner cells therefore receive multiple sequential corrections, with primitive recovery, floors, and EOS calls after each one. If a direction has only one interior cell, both sides update that same cell. These operations should be read as sequential updates of the current state, not as a single combined flux correction followed by one floor/EOS step.

7 Auxiliary arithmetic and scope

The routine also computes an unused quantity

$$\text{dhdt} = \frac{\min_{1 \leq d \leq \text{NDIM}} h_d}{\Delta t}. \quad (26)$$

For guard-cell index bounds ℓ_d^{GC} and u_d^{GC} , the storage extents and work-array lengths are

$$n_d = u_d^{\text{GC}} - \ell_d^{\text{GC}} + 1, \quad (27)$$

$$n_{\text{massFraction}} = \text{NSPECIES} \max_{1 \leq d \leq \text{MDIM}} n_d, \quad (28)$$

$$n_{\text{eosData}} = \text{EOS_NUM} \max_{1 \leq d \leq \text{MDIM}} n_d. \quad (29)$$

Although gravity and GLM damping parameters are imported, this file contains no explicit gravity correction or additional damping formula. Its comment says parabolic damping is applied to the flux-correction difference; the executable operation here is the conserved-vector update and direct recovery of ψ' above.